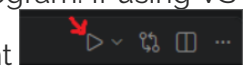
Lab Procedure

Play

Introduction/Setup

1. It's recommended that you run this lab in teams of at least 2 students
2. Move the QArm manipulator to the home position as outlined in the Concept Review, and turn ON the unit using the power switch located on the rear side of the base. Once powered, the manipulator should hold this position.
3. Open the "hardware" folder. This guide will assume VS Code is used, but feel free to run the programs in your own way.

Position Mode

1. Open **play_position_mode.py** in VS Code or a text editor of your choice.
2. Run the Program. If using VS code open the python file and press the “run” button at the top right 
    a. The Base light should turn green
    b. The QArm should move to the home position
    c. Two GUI windows as shown below will appear:”

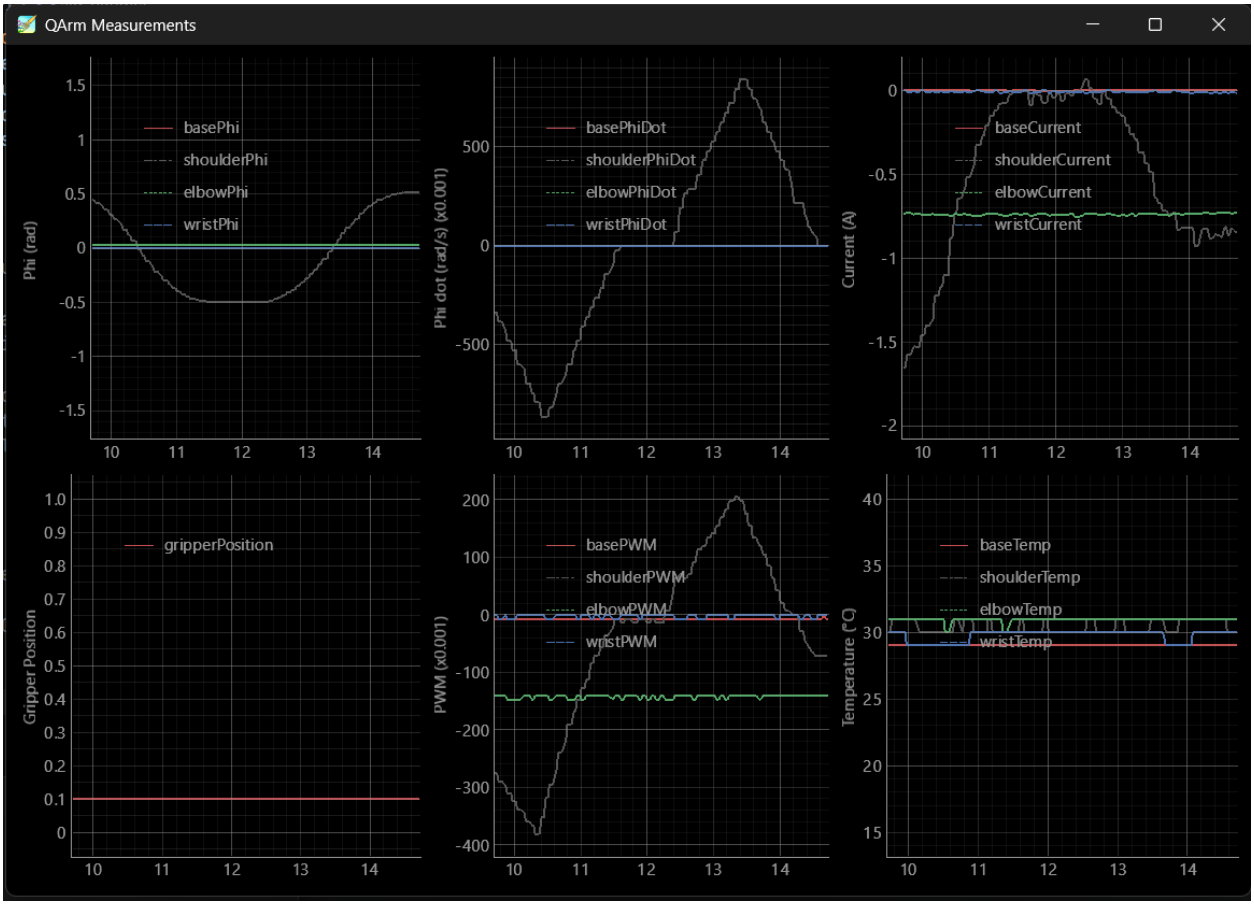


Figure 1: “QArm Measurements” window, which displays info about the arm

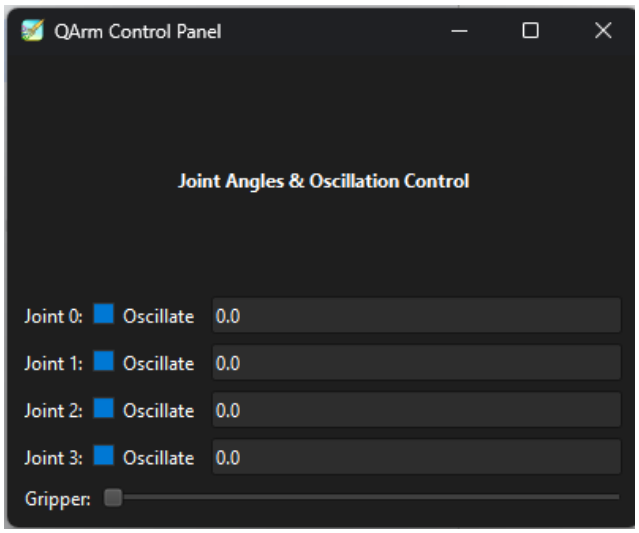


Figure 2: "QArm Control Panel" window, which will allow you to adjust values

3. Using the "QArm Control Panel" window, (Figure 2) adjust the joint positions by replacing the zeros with a number of your choice. This represents the joint angle in radians.

    a. Command small changes (i.e. ±0.25 rad) to determine the positive/negative angle convention

4. Using the gripper slider, adjust the position of the gripper. Under the hood, this is a value that ranges from 0 (open) to 1 (closed)

5. Toggle the oscillate checkbox to send a square wave signal to the desired joint. This will cause the joint to oscillate between +0.5 rad to −0.5 rad with a period of 6 seconds (3 seconds between each movement)

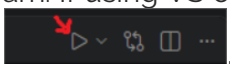
6. For all actions, use the virtual QArm visual and the "QArm measurements" window (Figure 1) to verify behaviour.

7. If no further experiment is required, close both GUI windows to end the program. If no more experiment is to be done, power off the QArm, otherwise, continue to the next section (PWM mode).

PWM Mode

1. Open **play_pwm_mode.py** in VS Code or a text editor of your choice.

> ⚠ Caution: When run, the QArm will start in PWM mode. It will go limp and all joints will slowly "fall" based on gravity and any applied force. Ensure that one group member is supporting the QArm

2. If in a team, discuss who will hold the arm and who will run the program. You may alternate after to ensure everyone gets hands on experience.
3. The chosen student should now hold the arm and prepare to support it.
4. Run the Program. If using VS code open the python file and press the "run" button at the top right  . This will star the program (visible as the LED base turning green), placing the arm in a "limp" state where all joints are receiving 0V.
5. The member that is holding the QArm in place should gently ease the arm into a neutral position where no joints are near their limits – ideally the home position, and gently hold it in place.
6. A slightly different UI will appear.
    a. The "QArm measurements" window is the same and shows the same data, except for the gripper position scope will not show as it is not used in this exercise.
    b. A new Control window will appear.

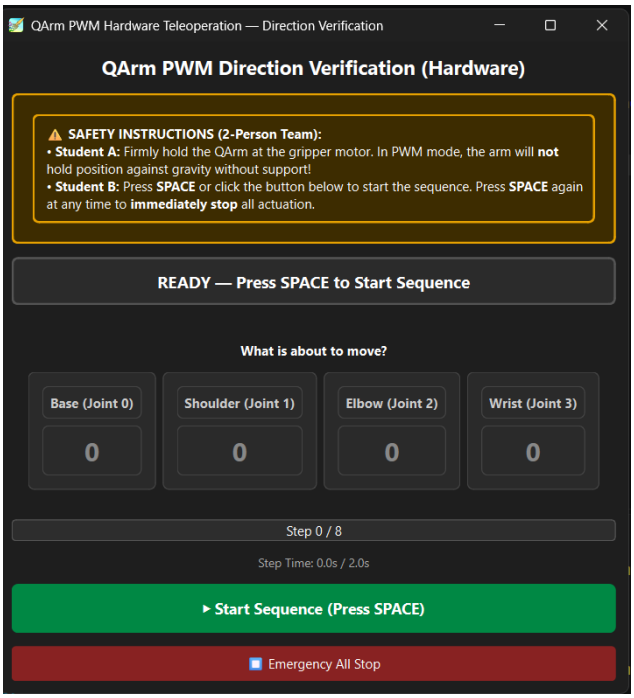


Figure 3: Control window for PWM mode

7. By pressing space or the green start button, the arm will now cycle through +1 and –1 command to each joint in order from base to wrist. This will get displayed in the Control window, and the student holding the hard will feel it moving/pushing. Communicate with your teammates to verify that the intended directions match. Also note the positive/negative directions and take notes to ensure they match those form the position mode lab.

8. Once the sequence is completed, the PWM signals will be reset to 0 and the arm will be fully limp again. Alternatively, the "Emergency All Stop" button will force the PWM signals to 0 if needed. Trade places and try again if needed.

9. The lab is now complete. Close both windows to terminate the python program. Turn OFF the manipulator using the switch in the read – make sure to catch the arm as it will fall when power is turned off – and gently move it to the rest position.